

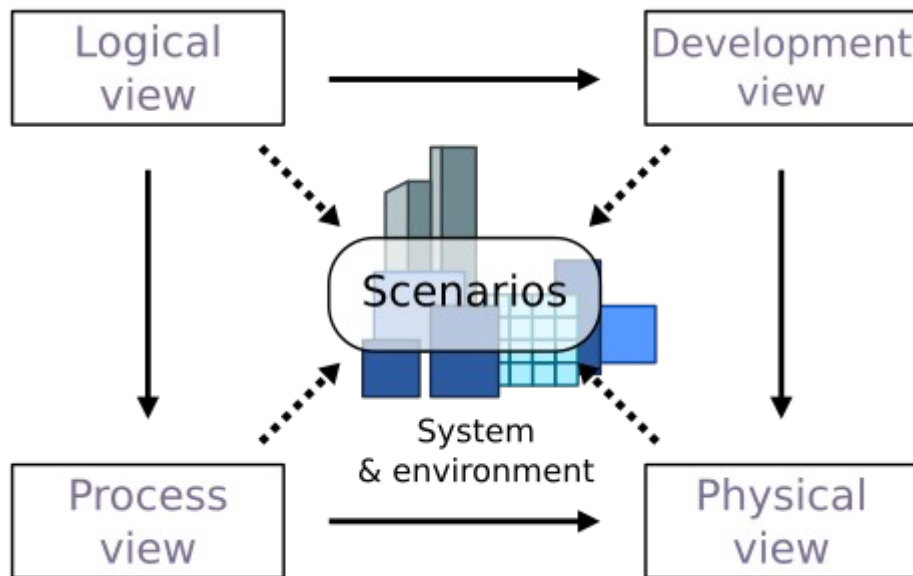
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1 Architectural view model

- 5 The architecture 4+1 views are used to describing the software systems based on different viewpoints, to discover if the architecture fulfills the needs of the different stakeholders. The views will be presented using the Unified Modeling Language (UML) tool



llibre

1.1 The logical view

- 10 The Logical View, often referred to as the “Class Diagram”. It focuses on the static structure of the system. This view deals with the essential components of the system, such as classes, objects, relationships, and their attributes. Class diagrams visually represent the classes and their associations in the system, to understand how various components interact with one another. This view provides insights into the overall design and organization of the system’s functionality.
- 15 Related UML diagrams:
- Class diagrams
 - Component diagram

1.2 The Physical view

- 20 It concerns the topology of the software components on the physical infrastructure as well as the physical connections between components. It includes details about servers, nodes, and their interconnections, including defining the communication protocols This view ensures that the system architecture aligns with the physical constraints and requirements.

Related UML diagrams:

- Class diagrams
- 25 • State diagrams
- Component diagrams

1.3 The Development view

30 The system from the perspective of the programmers (or the implementation view) and is concerned with software management. It maps the logical design to physical code organization, clarifies module boundaries, dependencies, and re-usability.

1.4 The Process view

The dynamic aspects of the system, explains the system processes and how they communicate. Models the dynamic behavior, execution flow, threading, synchronization, performance, and fault tolerance.

35 Related UML diagrams:

- Sequence diagrams,
- Activity diagrams
- State diagrams

1.5 The Scenarios (or use case view)

40 It captures what the system does from an end-user perspective. The description of an architecture is illustrated using a small set of use case, or scenarios, which become a fifth view. The scenarios describe sequences of interactions between objects and between processes. They are used to identify architectural elements and to illustrate and validate the architecture design.

Related UML diagrams:

- 45
- Use Case diagram

2 Architectural guidelines

2.1 Loosely coupled architecture

50 The architecture should follow the guidelines for loosely coupled architecture. A loosely coupled architecture is a design principle in which system components are weakly associated. This means that each component can operate independently, allowing for greater flexibility in system design and maintenance.

Key characteristics:

- 55
- Changes made to one component have minimal impact on others, which enhances the overall resilience of the system
 - Each component has little or no knowledge of the internal workings of other components.
 - Systems can scale more easily as components can be modified or replaced independently.
 - Each component can be developed, deployed, tested and maintained independently

60 2.2 Asynchronous component communication

A loosely coupled architecture using a queue allows different components of a system to communicate asynchronously, meaning they can operate independently without being directly connected.

65 The component communication uses the publish/subscriber model, in which a communication queue will be assigned to each of the subscribers. When a new message is published, the related message Topic will be distributed to the assigned subscribers queue.

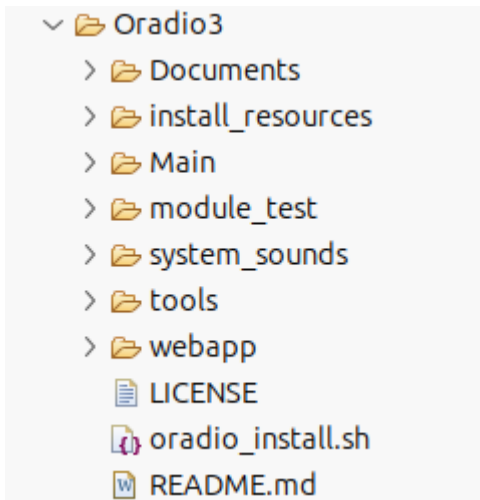
Key characteristics:

- Independence: Producers and consumers don't need to know about each other. They only interact with the queue, reducing direct dependencies.
- 70 • Failure isolation: If one component fails, the queue allows other components to continue functioning. This isolation minimizes the impact of failures on the overall system.
- Asynchronous Communication: Components can communicate without waiting for each other to respond, which reduces downtime and improves responsiveness.
- 75 • Buffering: They act as buffers that store messages until the receiving service is ready to process them. This helps manage varying loads and prevents service overload

3 The Deployment view

3.1 Software artifacts

3.1.1 Folder structure



The folder structure of the Oradio project :

- Documents: for design documentation
- install_resources: resources related to configuration files, system service file, shell scripts required for installation of Oradio project.
- Main: the python modules
- module_test: module test python programs
- system_sounds: sound files (wav) for system related sound (e.g. click.wav for touch button click sound)
- tools: shell scripts
- webapp: resources required for the web-interface.
- LICENSE: The GNU general public license, which is required to include to inform users of their rights and obligations regarding the software. This ensures proper attribution and compliance with the terms of the license.
- oradio_install.sh: install script to install a fresh Oradio project from scratch
- README.md: install instructions

3.1.2 Github

GitHub is a cloud-based platform for hosting and managing code that allows developers to collaborate on projects using version control. It is primarily used for storing code, tracking changes, and facilitating teamwork on software development projects. In this project github will be used for:

- Version Control: to track changes in code over time.
- Repositories: to store code, enabling sharing of code and collaboration
- Pull Requests: method to propose changes to a project, allowing for code review and discussion before integration.
- Issue tracking: to track bugs and feature requests, to manage the projects workflow.

3.1.3 Music sources

The Oradio is a music player that can play music from different sources:

- Music that is provided on the USB memory device.
- A music stream from a web-radio
- A music stream from Spotify player application

3.1.4 Self-documenting python code.

Module headers should have following Google style docstring formatting:

```
"""
    #####          ##          #####          #          #####
    #   #   #   #   #   #   #   #   #   #   #   #   #
    #   #   #   #   #   #   #   #   #   #   #   #
    #   #   #####   #####   #   #   #   #   #   #
    #   #   #   #   #   #   #   #   #   #   #   #
    #####   #   #   #   #   #####   #   #####

Created on May 28, 2026
@author:      Henk Stevens & Olaf Mastenbroek & Onno Janssen
@copyright:   Copyright, Oradio Stichting
@license:     GNU General Public License (GPL)
@organization: Oradio Stichting
@version:     1
@email:       oradioinfo@stichtingoradio.nl
@status:      Development
@summary:     One line summary of the module
@description: Detailed description of module
"""
```

- @author: the author(s)
- @copyright, @license, @organization, @version, @email, @status: open-source related information.
- @summary: one-line summary of module
- @description: detailed description of modules
- For each class, method or function there should be Google style documentation strings (docstring) , with following format:

```
def example_function(param1, param2):
    """Summary line.

    Detailed description of the function's purpose and behavior.

    Args:
        param1 (int): Description of the first parameter.
        param2 (str): Description of the second parameter.

    Returns:
        bool: True if successful, False otherwise.

    Raises:
        ValueError: If param1 is negative.

    Examples:
        >>> example_function(1, 'test')
        True
    """
```

- **Summary line:** A brief description of the module, class, or function, limited to one line.
- **Detailed Description:** This section provides a more comprehensive explanation of the code's purpose and usage.
- **Sections:**

Section	Description
Args	Lists the parameters accepted by the function, including their types.
Returns	Describes the return value and its type.
Raises	Lists exceptions that the function may raise, along with conditions.
Examples	Provides usage examples to illustrate how to use the function or class.

3.2 Target environment

- The RaspberryPi-3A+ headless module will be used as central control unit, providing hardware for:
 - Connectivity: WiFi, Bluetooth, Ethernet, USB
 - Input/Output pinning (GPIO): for user interface, as buttons and led indications
 - RaspiOS: Raspberry Pi OS (open-source) is a Unix-like operating system developed for the Raspberry Pi line of single-board computers. Based on Debian, a Linux distribution, it is maintained by Raspberry Pi Holdings and optimized for the Pi's hardware, with low memory requirements and support for both 32-bit and 64-bit architectures.
 - Extension interface at GPIO for Audio Playback (DigiAmp), touch buttons, volume knob and led drivers.
- Raspberry Pi OS: latest version, after being tested by developers

- Python is the programming environment, which has a wide range of libraries specifically designed for hardware interaction, created by other developers, eliminating to write everything from scratch. Comes preinstalled on the Raspi-OS

145 **3.3 Software installation**

3.3.1 Development: install script

The install shell script will install and configure the environment for the Oradio:

- logging: create logging folders
- services: install services, using the service templates in install_resources folder. Enable and start these services.
- Linux packages: install required Linux packages
- Python packages: install required Python packages
- Configuration: adapt configuration files

150

3.3.2 Production

For production an image is created from a released software version. The “scrum-master” is responsible for handing over an image for production.

155

3.4 Runtime

- runs on boot
- logging
- handling errors

160

3.5 Configuration

- Configuration items:
 - config.txt,
 - asound.conf,
 - mpd.conf,
 - logrotate.conf
- ??

165

3.6 System services

- systemd services:
 - librespot.service

170

- mpd.service
- oradio.service
- usb_low_idle_power_service: (still needed??)
- Dynamic device management: Udev rules for USB devices

175

3.7 Plugins

The Oradio should be prepared for adding new functionalities without or with minimal consequences for the basic Oradio. The basic Oradio offers following services:

- Touch button handling
- 180 • Led indication handling
- System sound for announcements
- Internet connectivity (WiFi)
- Web-interface to configure or customize the Oradio
- USB support to import music files and playlists
- 185 • Music player: Music Player Daemon (MPD)

The basic state-machine is setup to handle above listed services. The objective is to maintain the python code as a one-code-base. Additional services should be offered as a plugin, enabling dynamically loading and executing the new functionality, without affecting the basic state-machine. Example for plugins:

- 190 • Spotify streaming services
- Bluetooth speaker services
- Presence sensor services
- Bluetooth based light control.

The plugins module should be available in the code base. Via the web-interface a plugin can be enabled/disabled.

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Python based plugins require a plugin framework with following elements:

- a project structure for the plugins
- a plugin class (or interface)
- a loader

200 • a life-cycle handler

3.7.1 Project structure

```
/Main
==> /plugins/
    plugin_module_1.py
    plugin_module_2.py
    plugin_module_n.py
==> /core/
    plugin_loader.py
    oradio_control.py
    touch_buttons.py
    etc
```

The plugins directory holds the plugins modules. The plugin_loader is in the /core folder, which provides a base class (or interface) that all plugins must implement, to ensure consistency.

3.7.2 Plugin base class

205 The base class should provide following methods:

class PluginBase:

def initialize(self,config:dict)

Called once when plugin is loaded, to setup the plugin

@abstractmethod

210 def run(self)

Main logic of the plugin, called by the framework to execute the plugin.

def cleanup(self)

Called when the plugin is unloaded or shuts down, to cleanup resources

def on_event(self, event:str)

215 Called upon an event to be processed by main state machine

def health(self) → HealthStatus

returns the health status of plugin

def on_error(exc:BaseException)

220 if exceptions arise around the plugin execution, route them to the method, for further processing

Each plugin should inherit from this class and implement the methods

3.7.3 Dynamic Plugin loading

Use Python's <importlib> to dynamically load plugins from plugins directory

```
import importlib
import os
from pathlib import Path

def load_plugins(plugin_dir="plugins"):
    plugins = []
    for file in Path(plugin_dir).glob("*.py"):
        module_name = file.stem
        module = importlib.import_module(f"{plugin_dir}.{module_name}")
        if hasattr(module, "plugin") and isinstance(module.plugin, Plugin):
            plugins.append(module.plugin)
    return plugins
```

225 **3.7.4 life-cycle handler**

The life-cycle handler could be organized in a state-machine, managing the plugins for loading, initialization, running and cleanup, from one place. In case a plugin crashes, the handler can catch the error and continue running the other plugins.

230 Important is that a plugin runs in its own environment, e.g. as a Thread. This allows plugins to run in parallel and a Thread can be stopped if needed with a stop signal

See also the [PluginManager State control section](#)

4 The Logical view: Class diagrams

- 235 The Oradio3 software layering typically consists of three main layers:
- the **hardware abstraction layer** (HAL), which includes the physical components.
 - the **middle-ware layer**, which facilitates communication and data management between the hardware and applications. The OS services is considered to be part of this layer
 - the **application layer**, where the actual software applications run and perform specific tasks.
- 240 • the **globals&utils layer**, which provide global and public functions

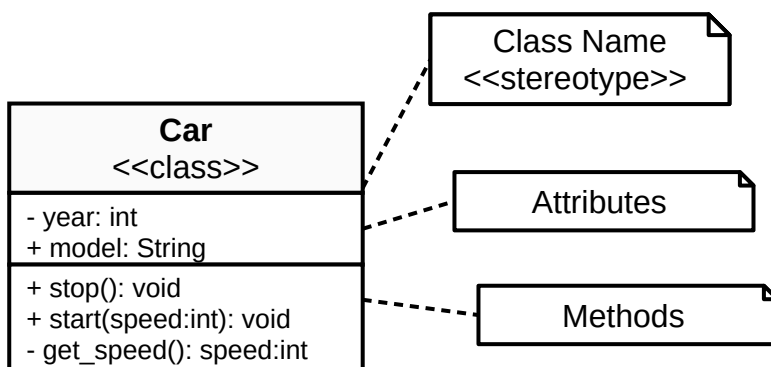
The diagrams are organized from a top-level view and for each python module the class design.

4.1 Introduction on Class diagrams

- 245 UML class diagrams are visual representations of the structure of a system, showing classes, their attributes, operations, and the relationships among objects. Class diagrams helps in modeling object-oriented systems and understanding how different components interact within the software.

4.1.1 UML class notation

Classes are depicted as boxes, each containing three compartments for the class name, attributes, and methods.



- **Class Name:** The name of the class is written in the top compartment of the class box and is centered and bold. The name may contain the name of the module in which the class is defined. E.g. <<class>>log_service.SafeLogger the class SafeLogger is defined in the module log_service. The <<stereotypes>> are used to extend the semantic of UML elements, providing additional information about the class. Examples of stereotypes:

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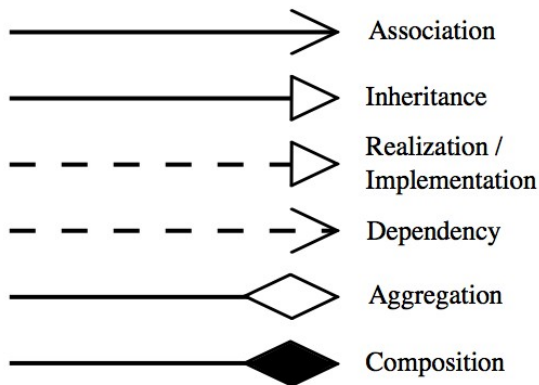
 - <<singleton>> : single instance class
 - <<dataclass>>: data holder class
 - <<static>>: a public class
 - <<module>>: name of the (Python) module
- **Attributes:** Attributes, also known as properties or fields, represent the data members of the class. They are listed in the second compartment of the class box and often include the visibility (e.g., (+) public, (-) private) and the data type of each attribute.
- **Methods:** Methods, also known as functions or operations, represent the behavior or functionality of the class. They are listed in the third compartment of the class box and

260

265 include the visibility (e.g., (+) public, (-) private), return type, and parameters of each method.

4.1.2 UML Class relationships

In class diagrams, relationships between classes describe how classes are connected or interact with each other within a system. Here are some common types of relationships in class diagrams:

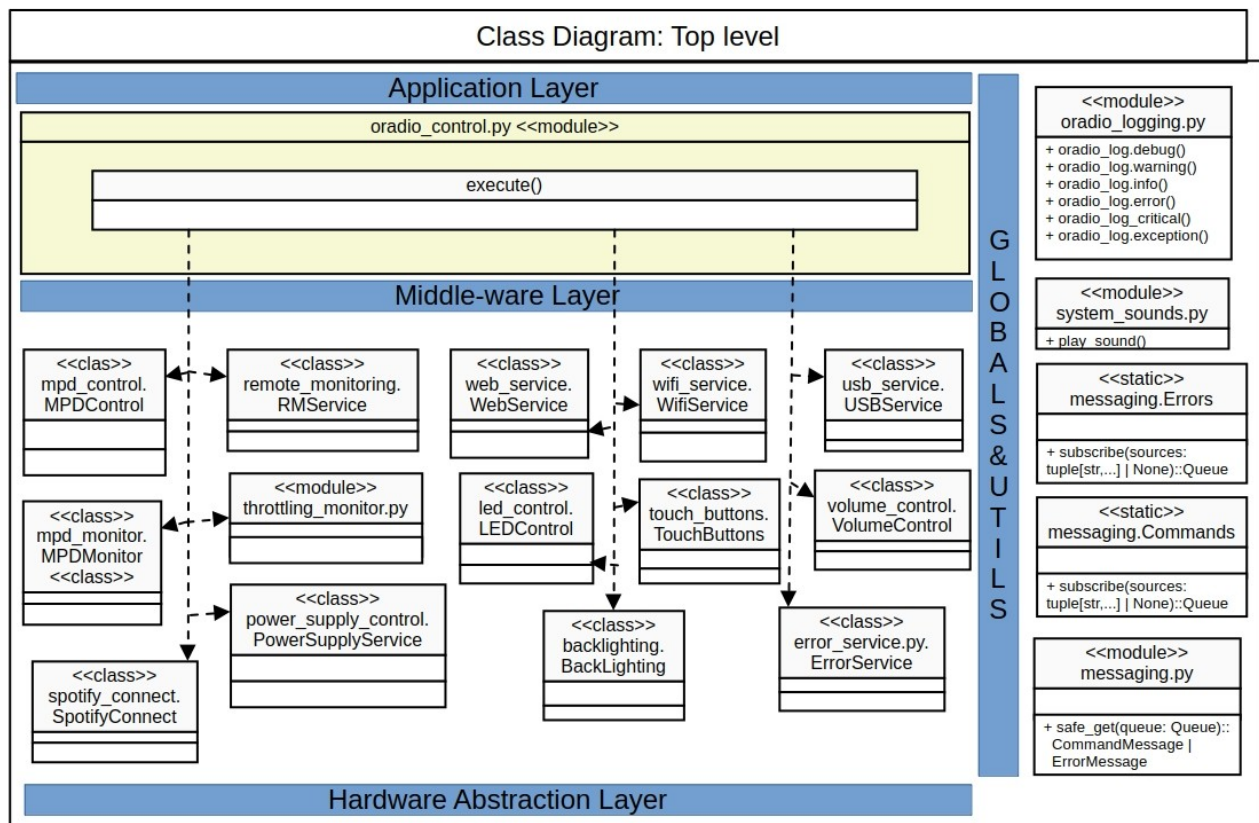


- **Association:** An association represents a bi-directional relationship between two classes. It indicates that instances of one class are connected to instances of another class. Associations are typically depicted as a solid line connecting the classes, with optional arrows indicating the direction of the relationship.
- 275 • **Inheritance:** Inheritance represents an "is-a" relationship between classes, where one class (the subclass or child) inherits the properties and behaviors of another class (the superclass or parent). The subclass inherits all the functions of the parent class, and the parent class has all the attributes, methods, and subclasses
- 280 • **Realization/Implementation:** Realization indicates that a class implements the features of an interface. It is often used in cases where a class realizes the operations defined by an interface.
- 285 • **Dependency:** A usage dependency ("uses") relationship in a UML class diagram indicates that one class (the client) utilizes or depends on another class (the supplier) to perform certain tasks or access certain functionality. The client class relies on the services provided by the supplier class but does not own or create instances of it.
- **Aggregation:** Aggregation is a specialized form of association that represents a "whole-part" relationship ("has"). It denotes a stronger relationship where one class (the whole) contains or is composed of another class (the part). In this kind of relationship, the child class can exist independently of its parent class
- 290 • **Composition:** Composition is a stronger form of aggregation, indicating a more significant ownership or dependency relationship ("consists of"). In composition, the part class cannot exist independently of the whole class.

4.2 The top level diagram

There is one application running, being the python module `oradio_control.py`, which will initialize and create all instances necessary for the operation. The `execute()` function indicates that

300 `oradio_control.py` is executed and running.



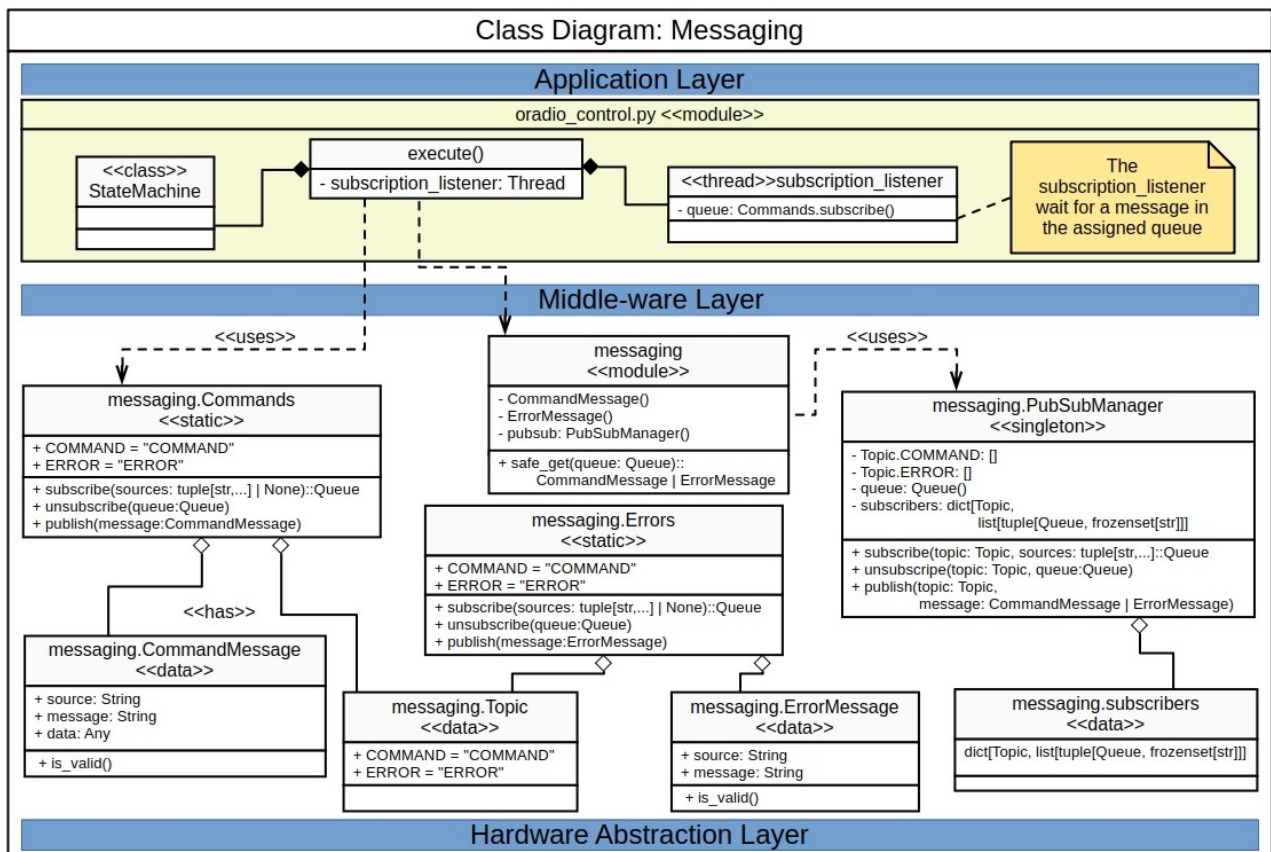
Most of the classes (stereotype `<<class>>`) will be instantiated during initialization phase of execution.

The throttled_monitor is globally instantiated, a monitoring thread will be started, to check the condition of the raspberrypi.

The global methods or functions are represented in the GLOBAL&UTILS layer, as these functions are public, being:

- `oradio_log.xx()`
- `play_sound()`
- `subscribe_commands()`, `subscribe_errors()`, `publish_command()`, `publish_error()`

4.3 Messaging



The messaging.py module provides services for the publish-subscribe model as inter-module communication. It is a messaging pattern where publishers send messages to a topic without knowing who the subscribers are, while subscribers express interest in specific topics to receive those messages. This model allows for asynchronous communication and decouples the components, enhancing scalability and flexibility in distributed systems.

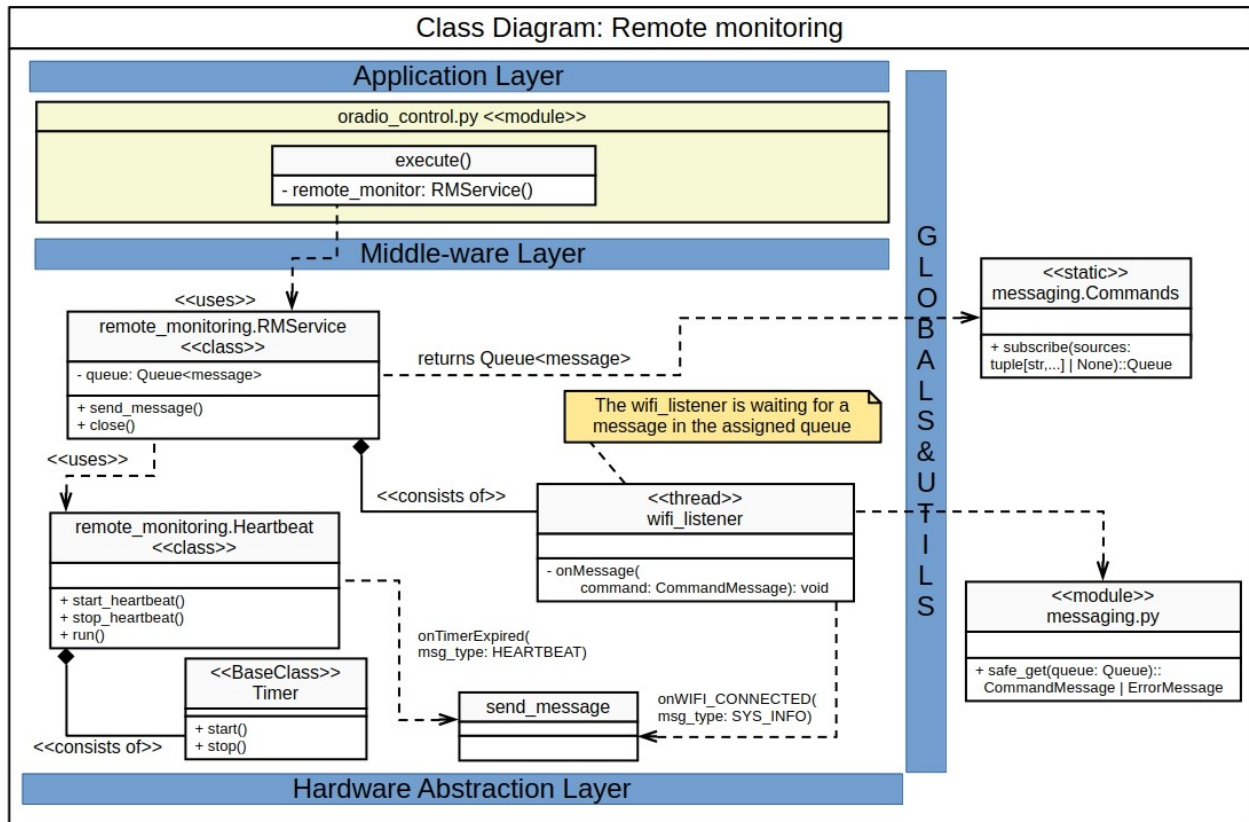
The module has a public API to subscribe to a certain Topic or publish a message to a Topic. The defined Topics are:

- COMMAND: representing a command
- ERROR: representing an error code

The public API provides following methods:

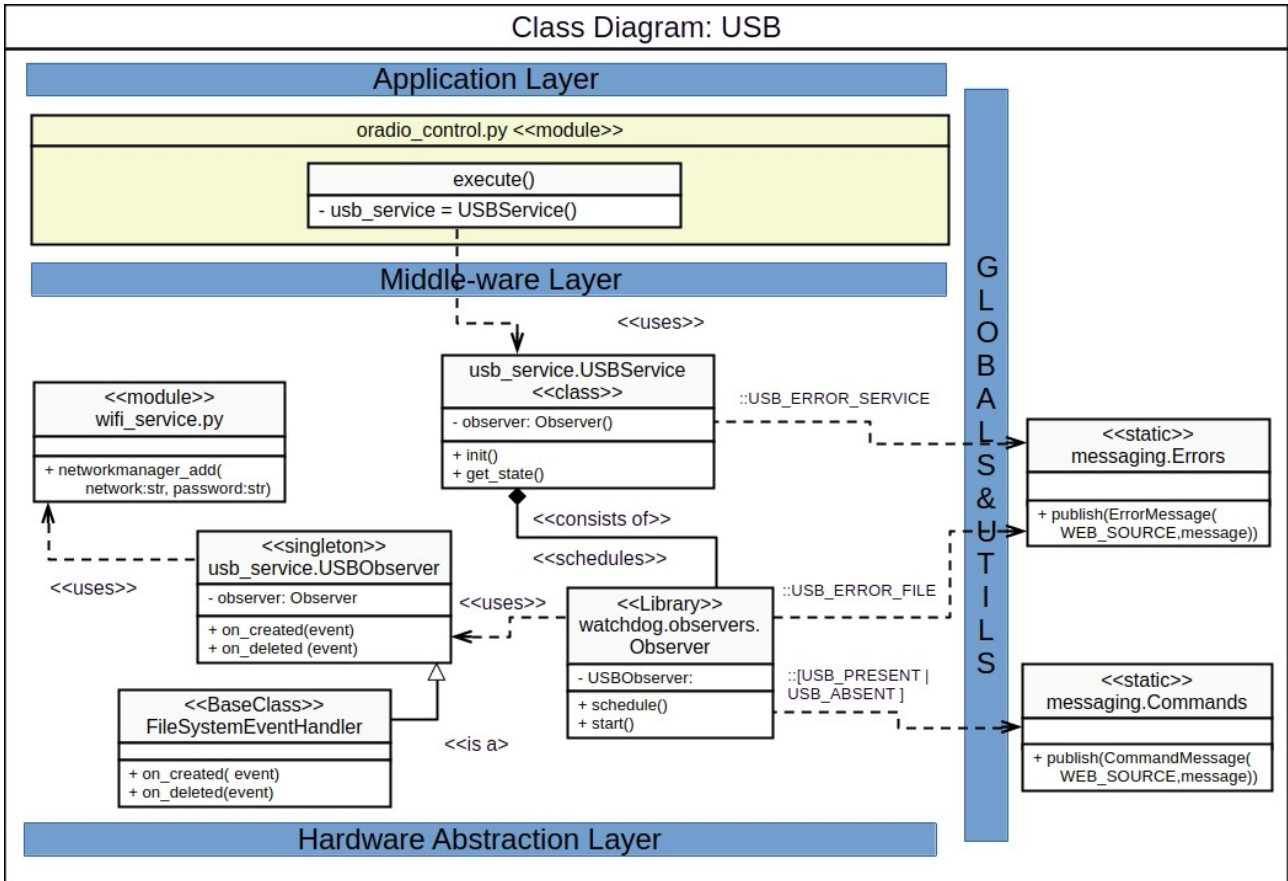
- subscribe_commands(): subscribe to Message Topic, returning a message queue to monitor for new commands
- publish_command(message: CommandMessage): publish a command to COMMAND Topic
- subscribe_errors(): subscribe to Error Topic, returning a message queue to monitor for new errors
- publish_error(message: ErrorMessage): publish a command to ERROR Topic

4.4 Remote Monitoring Service

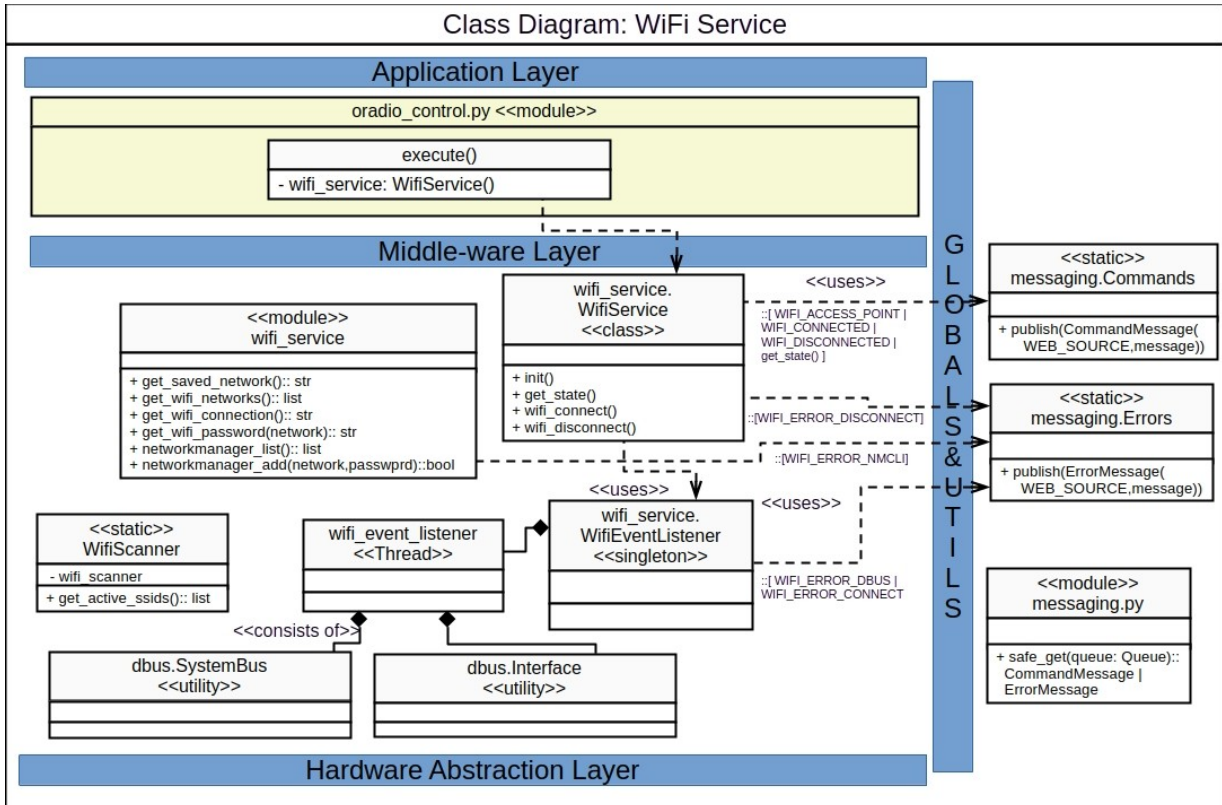


The Remote Monitoring Service maintains a connection with a Remote Server via Wifi. The class `RMService()` will be instantiated during initialization of `oradio_control.py`. The `RMService` class is depending on whether there is a WiFi connection available. So it initiates a thread `<wifi_listener>` that will listen to WiFi related commands, submitting a `subscribe_commands()`, which returns a message Queue. When WiFi is available it will send a `SYS_INFO` message to the server. A regular `HEARTBEAT` message is sent to server as a keep-alive indication for the server.

4.5 USB Service



4.6 Wifi Service



4.7 Logging

Oradio's logging module provides services for logging across all modules with safe, concurrent, and asynchronous capabilities. It is based on the Python's built-in logging module for emitting log messages to track events and errors during execution.

350 It is designed to operate robustly in multi-threaded applications, Raspberry Pi hardware, and web applications (Uvicorn/FastAPI).

The logging outputs are configured in Handlers, such as:

- Console logging: StreamHandler
- File logging: FileHandler
- 355 • Syslog logging: SysLogHandler (send logs to syslog)
- Email logging: SMTPHandler (send log to email)
- Web logging: HTTPHandler
- Logging Server: SocketHandler

The oradio logging module supports console, syslog and file logging.

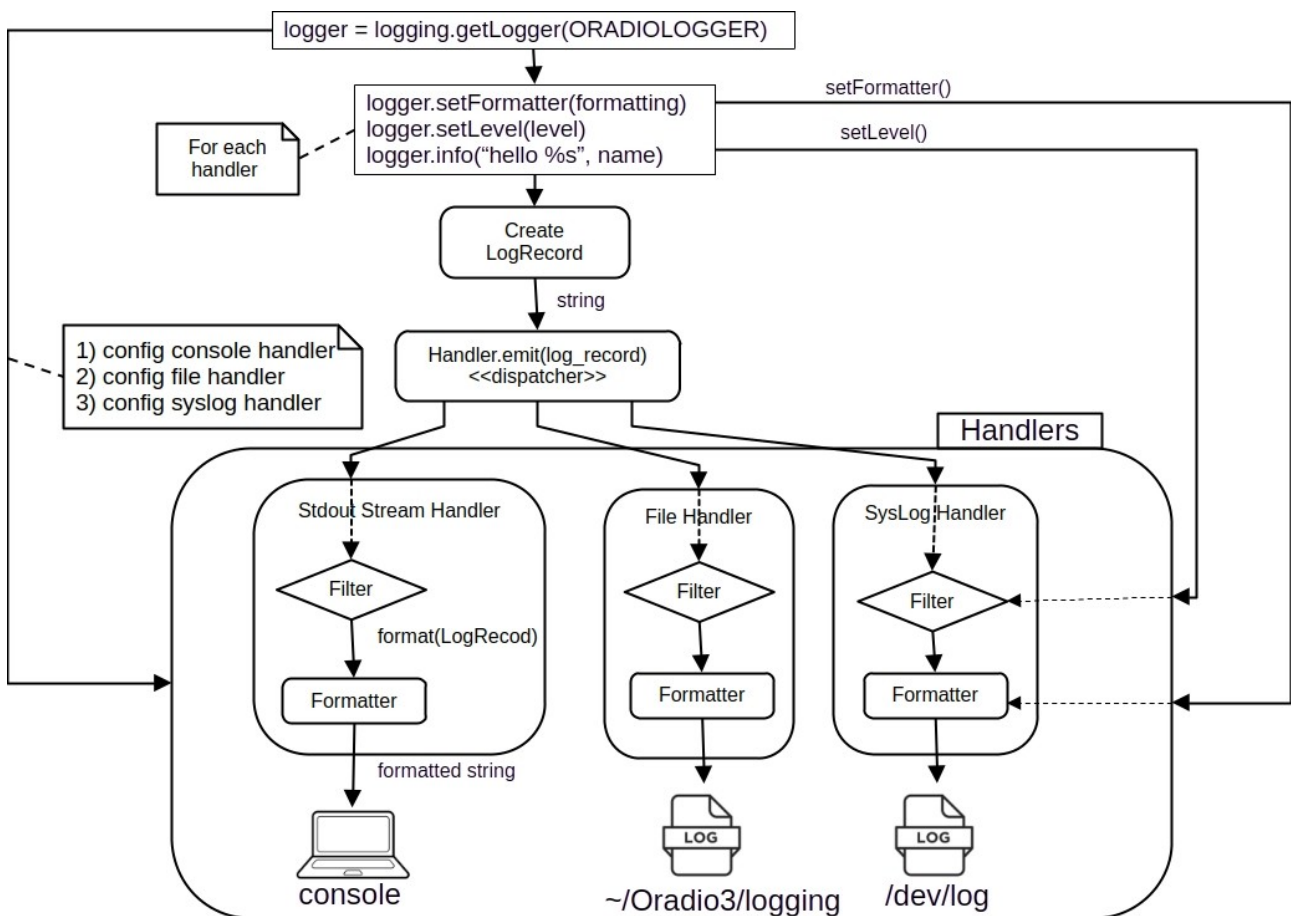


Figure 1: Logging activity diagram

A logger needs to be configured for its output. For Oradio 3 outputs are configured:

- Console output
- File output to `~/Oradio3/logging/oradio.log`. The FileHandler is a parent for a child class `ConcurrentRotatingFileHandler`, supporting size-based log rotation
- 365 • Syslog handler sends outputs to the system's central syslog server

4.7.1 Key features

- Thread-safe queue-based logging with asynchronous dispatch.

- Colored console output for better log readability.
- Concurrent file logging with rotation, safe across threads/processes.
- Custom TRACE log level (5) below DEBUG.
- Exception-safe handlers prevent recursive logging failures.
- Integration with Uvicorn/FastAPI by replacing default loggers.
- The Python interpreter runtime fault handler is enabled to capture crashes.
- Thread-safe queue-based logging.

4.7.2 Class diagram

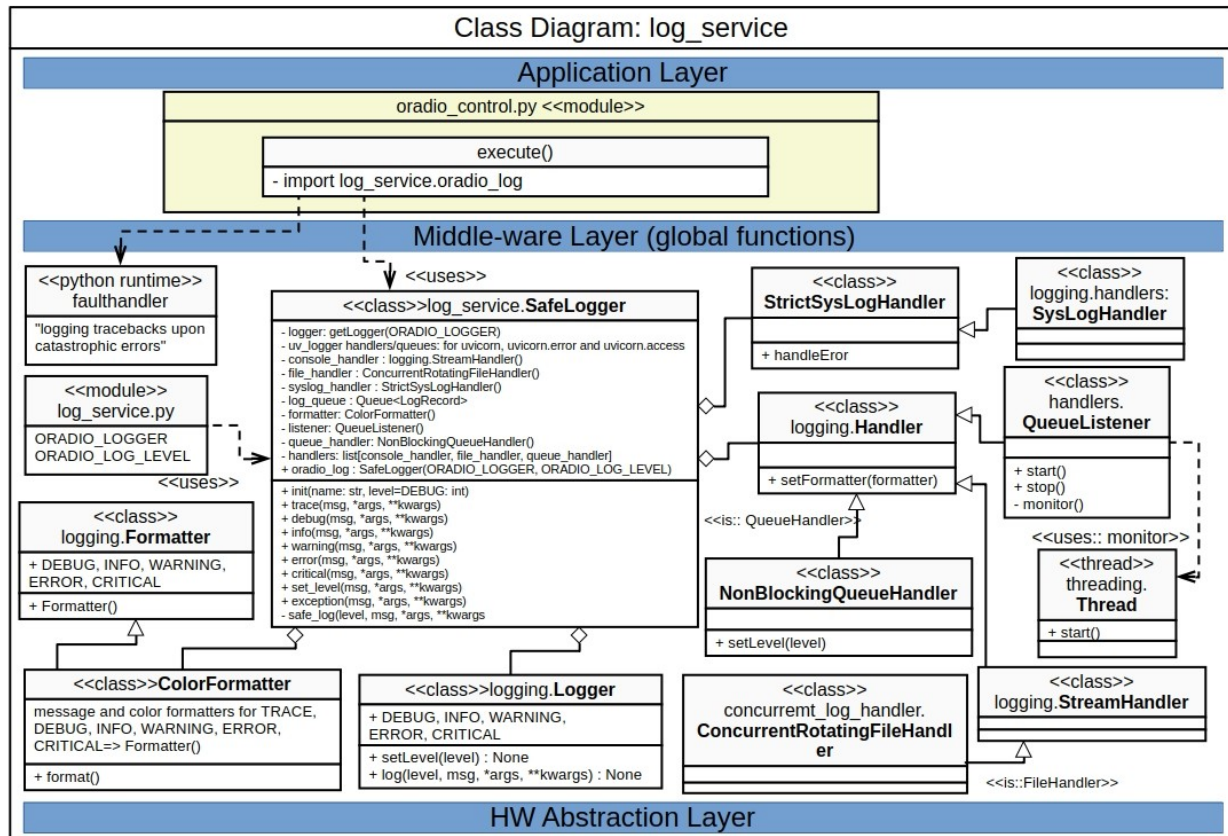


Figure 2: log_service class diagram

The public instance `oradio_log` of the class `SafeLogger` will be imported from `log_service` module during initial execution of `oradio_control.py` module. Also the fault handler will be enabled during initial execution.

4.7.2.1 fault handler

The Python fault handler is included during import of the logging module. When enabled it provides tools for diagnosing Python interpreter crashes and fatal signals. It will be registered to the Python Interpreter. When a catastrophic signal such as `SIGSEGV` or `SIGABRT` occurs, the fault handler prints a traceback to `sys.stderr` instead of leaving only an OS-level crash message. The `stderr` output is redirected to a file. It is implemented in C so it remains functional even when the interpreter is in a corrupted state.

4.7.2.2 *SafeLogger class*

390 All logging services are exposed under the static instance `<radio_log>` of the class `SafeLogger`. This static instance can be included on all modules which want to use logging. The `SafeLogger` is a wrapper on the python logging handler and exposes logging methods for: trace, debug, info, warning, error, critical, exception.

The `SafeLogger` is a aggregation of following classes:

- `StricySysLogHandler`
- `logging.Handler`
- 395 • `logging.Logger`
- `ColorFormatter`

4.7.2.3 *StrictLogHandler*

400 The `StrictLoghandler` (or child) inherits from the superclass `SysLogHandler`, and reuses all attributes and methods from superclass (or parent). In Python's logging framework, when `emit()` (or the handler's internal `send`) fails, the logging system usually calls `handleError()`. In the `standard logging.Handler` implementation, `handleError()` generally does not raise to the caller. It may print to `stderr` (depending on config) and then continue. That means your code can't reliably detect "syslog is broken" at startup. The `StrictLogHandler` subclass overrides

405 `handleError()` to raise an `OSError`.

4.7.2.4 *logging.Handler*

The `logging.Handler` provides several classes to manage the log message, such as:

- `StreamHandler`: Sends log messages to streams like `sys.stdout` or `sys.stderr`.
- `FileHandler`: Directs log messages to a specified file, allowing for persistent logging.
- 410 • `RotatingFileHandler`: Manages log files by rotating them after reaching a certain size, preventing file overflow.
- `QueueListener`: a component that consumes log records from a queue and dispatches to one of the log handlers

4.7.2.5 *logging.Logger*

415 Is the core class in Python's logging that represents a named logger (`ORADIO_LOGGER`), which can be used to emit log messages

4.7.2.6 *ColorFormatter*

420 The `<ColorFormatter>` is a specialization on the general `<logging.Formatter>`, that adds color styling to the log output. The `<logging.Formatter>` converts a logging message into a formatted text string. The formatting used is:

"radio[% (process)d]: % (filename)s: % (lineno)d - % (levelname)s - % (message)s"

- process: the numeric process id (PID) that emitted the log.
- filename: the base name of the file where the logging call occurred (not the full path)

- lineno: the line number in that file where the logging call occurred.
- levelname: the log level name (e.g., "DEBUG", "INFO", "WARNING", "ERROR", "CRITICAL").
- message: the final formatted log message text

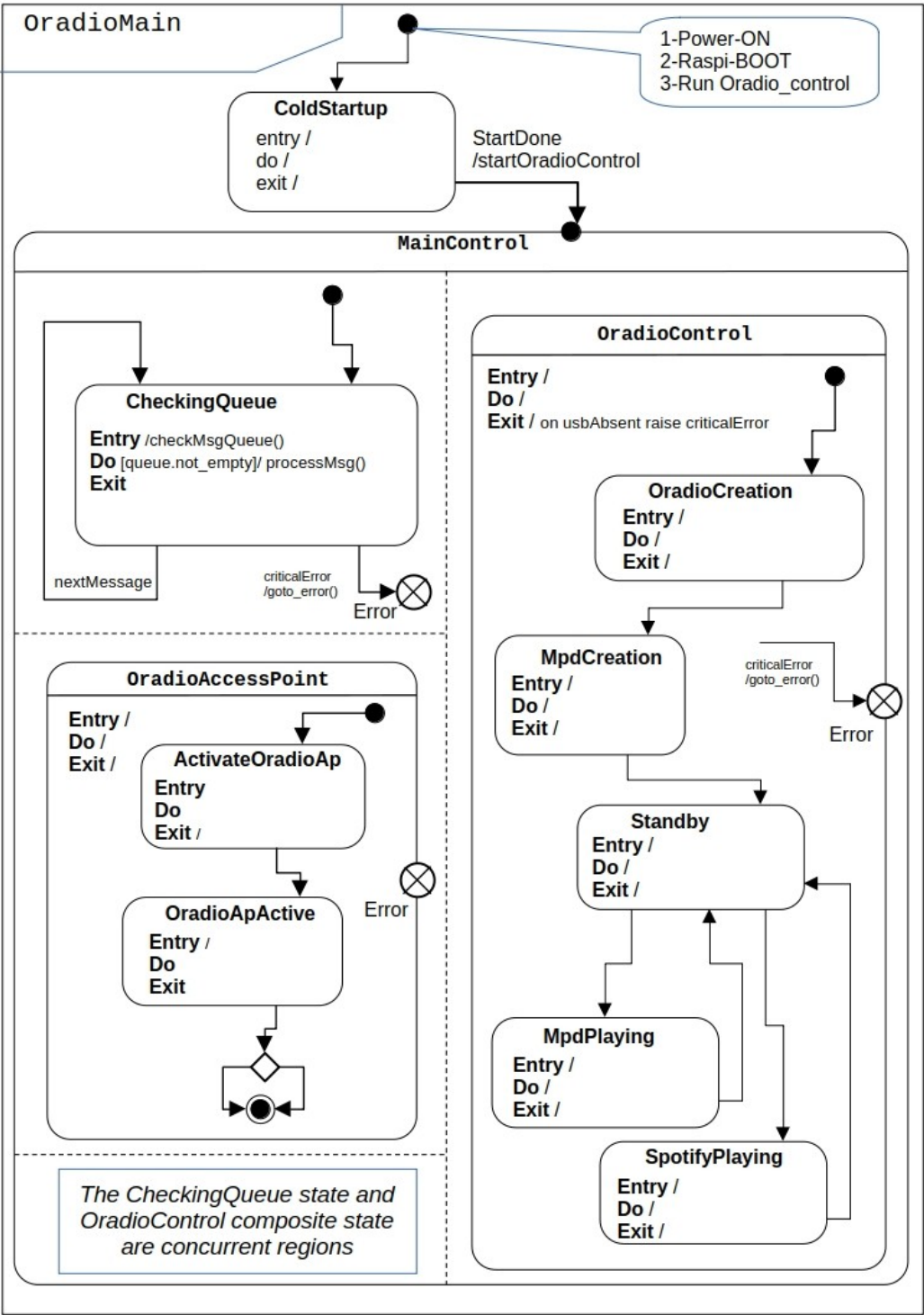
The <<ColorFormatter>> will add ANSI escape codes to the logging message text string. The following colors are defined:

Log level	Color
TRACE	Blue
DEBUG	Grey
INFO	White
WARNING	yellow
ERROR	Red
CRITICAL	Magenta

5 The Logical view: State diagrams

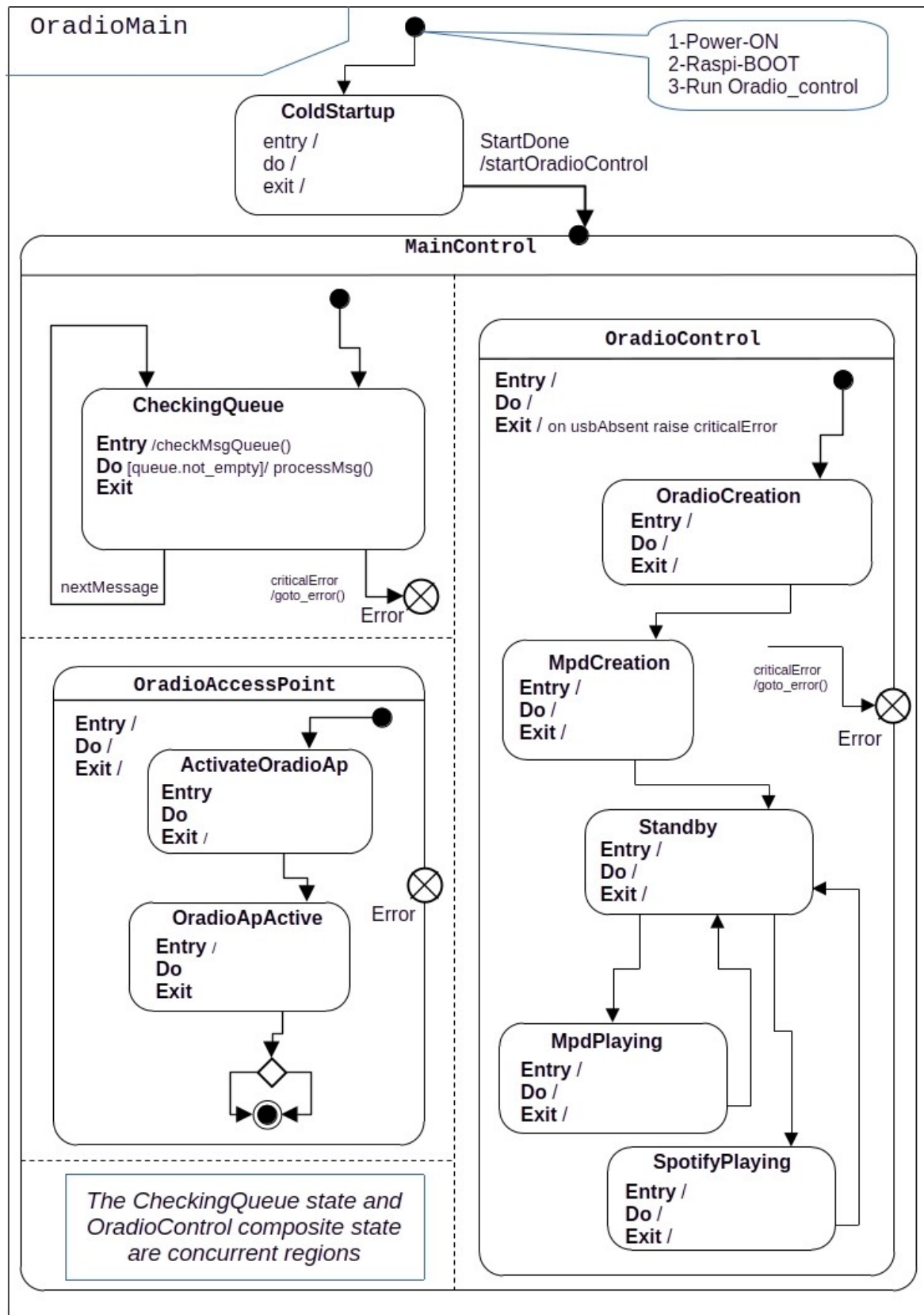
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5.1 Maincontrol state



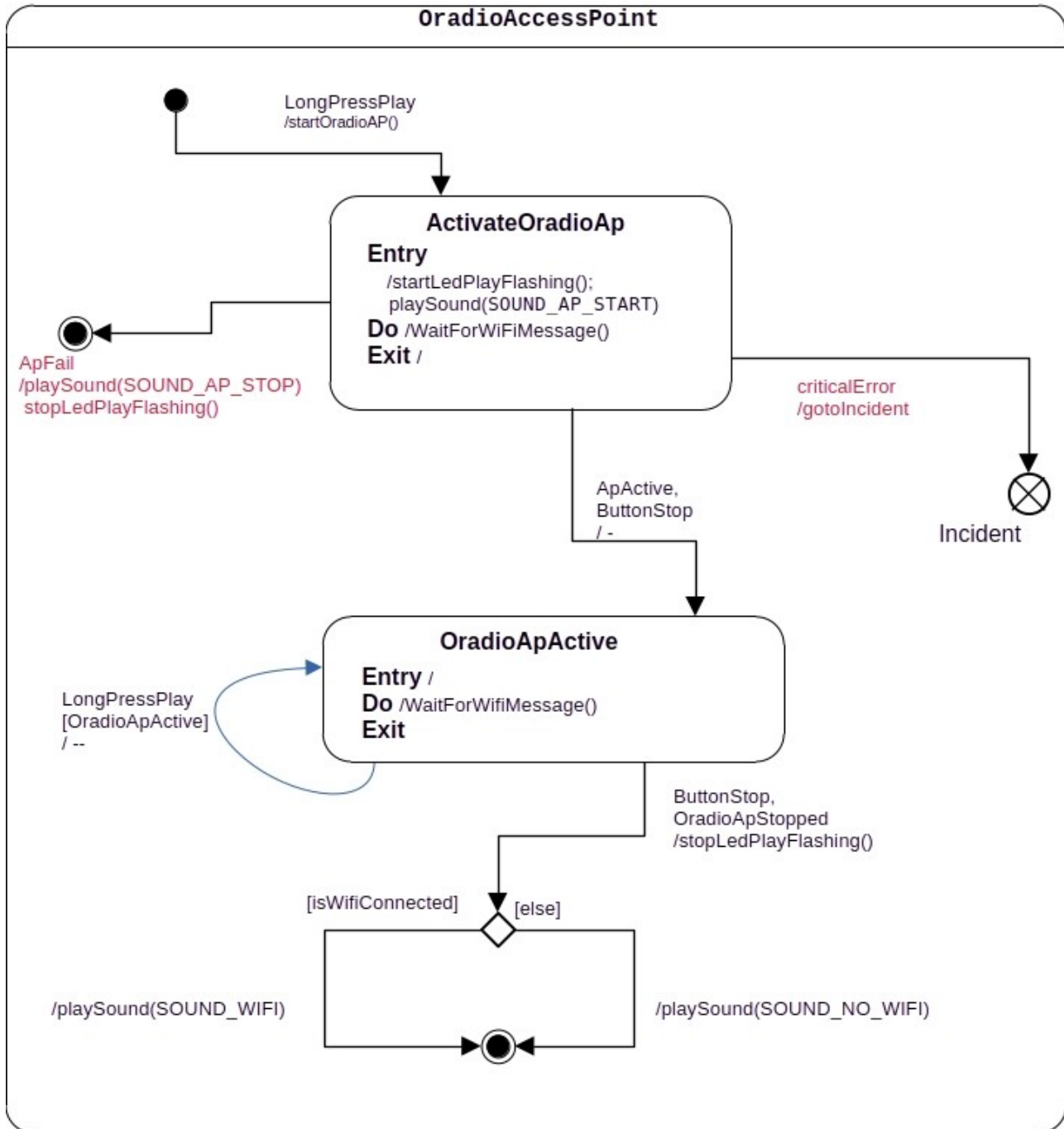
Description: xxxxxx

5.2 Oradio Control States

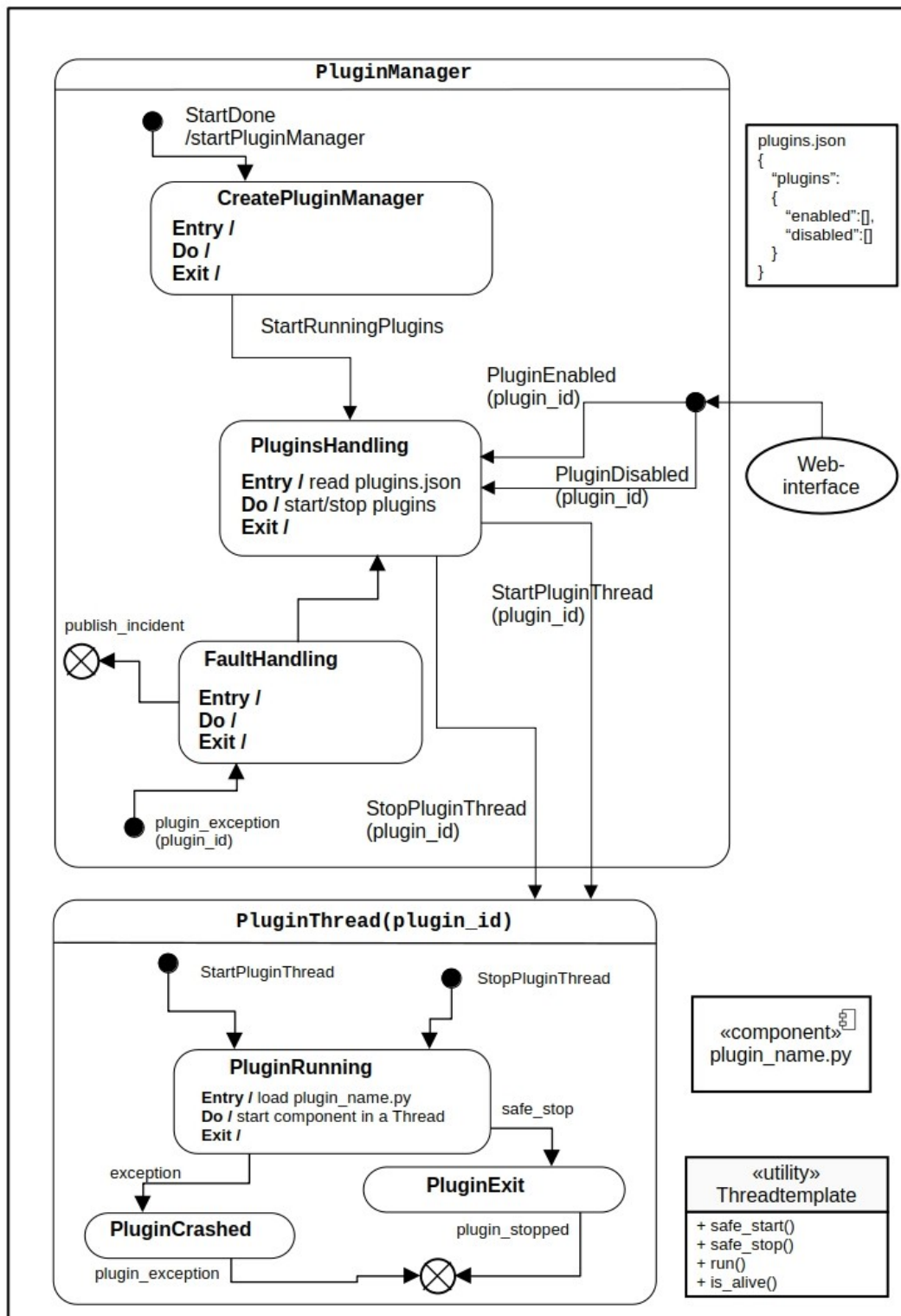


Description:xxx

5.3 Access Point state control



5.4 PluginManager State control



The plugins manager is responsible for controlling the plugins. The plugins are used to dynamical extend the functionality of the standard Oradio.

The standard Oradio is supporting:

- Touch button handling
- Led indications
- System sound for announcements
- Default music player: MPD
- Internet access via WiFi
- Web-interface to configure the Oradio

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As an example the Oradio could be extended with a Spotify service. This service will be offered as plugin. The plugins manager will start/run a plugin dynamically when it is enabled, as a Thread. So it each plugin runs in its own execution context. If a plugins has a fault the PluginsManager gets an exception, so the Thread can be removed.

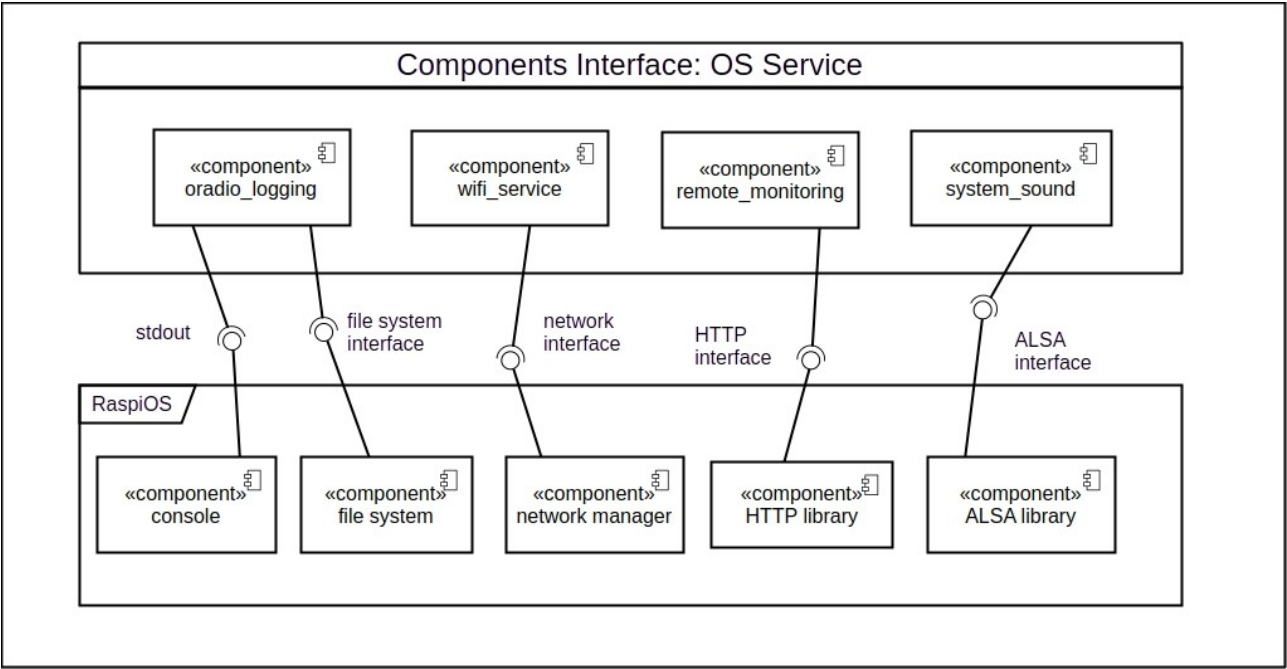
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The plugin framework provides an abstract base class for initialization, execution and cleanup of the plugin.

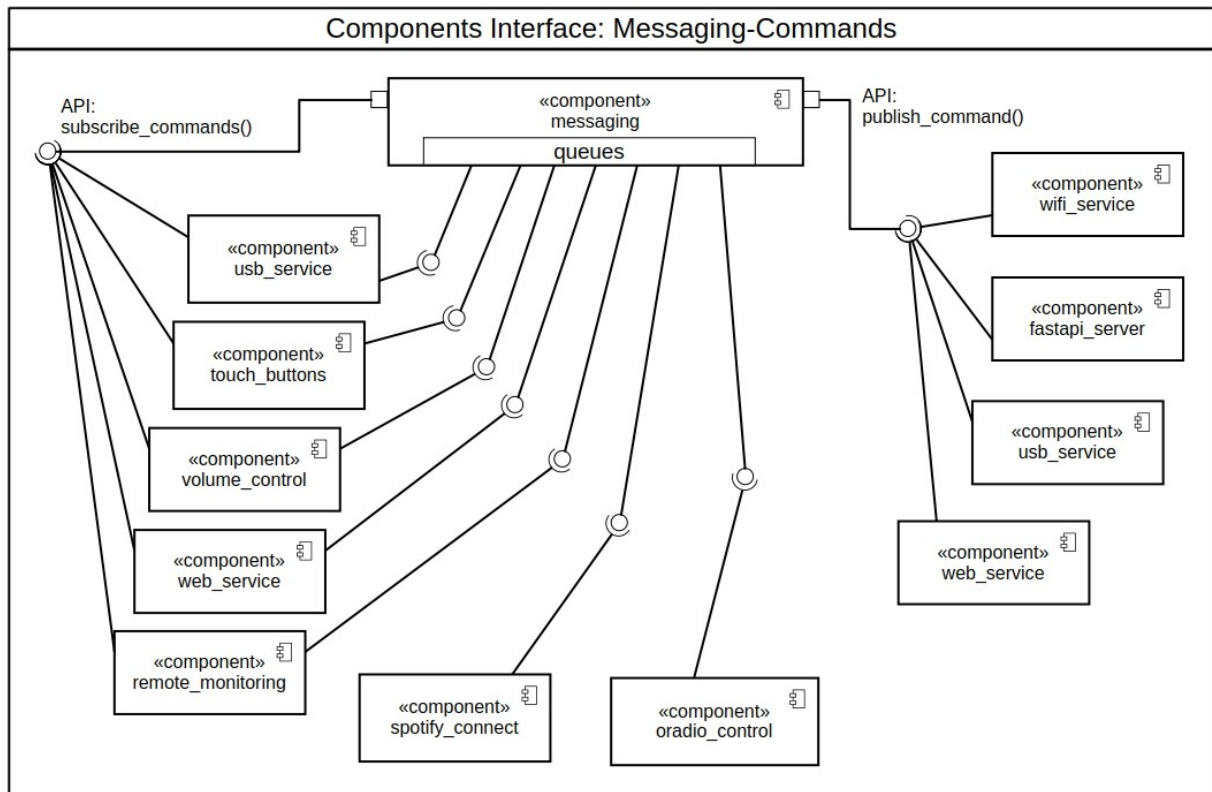
6 The Physical view: Component diagrams

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6.1 OS Service



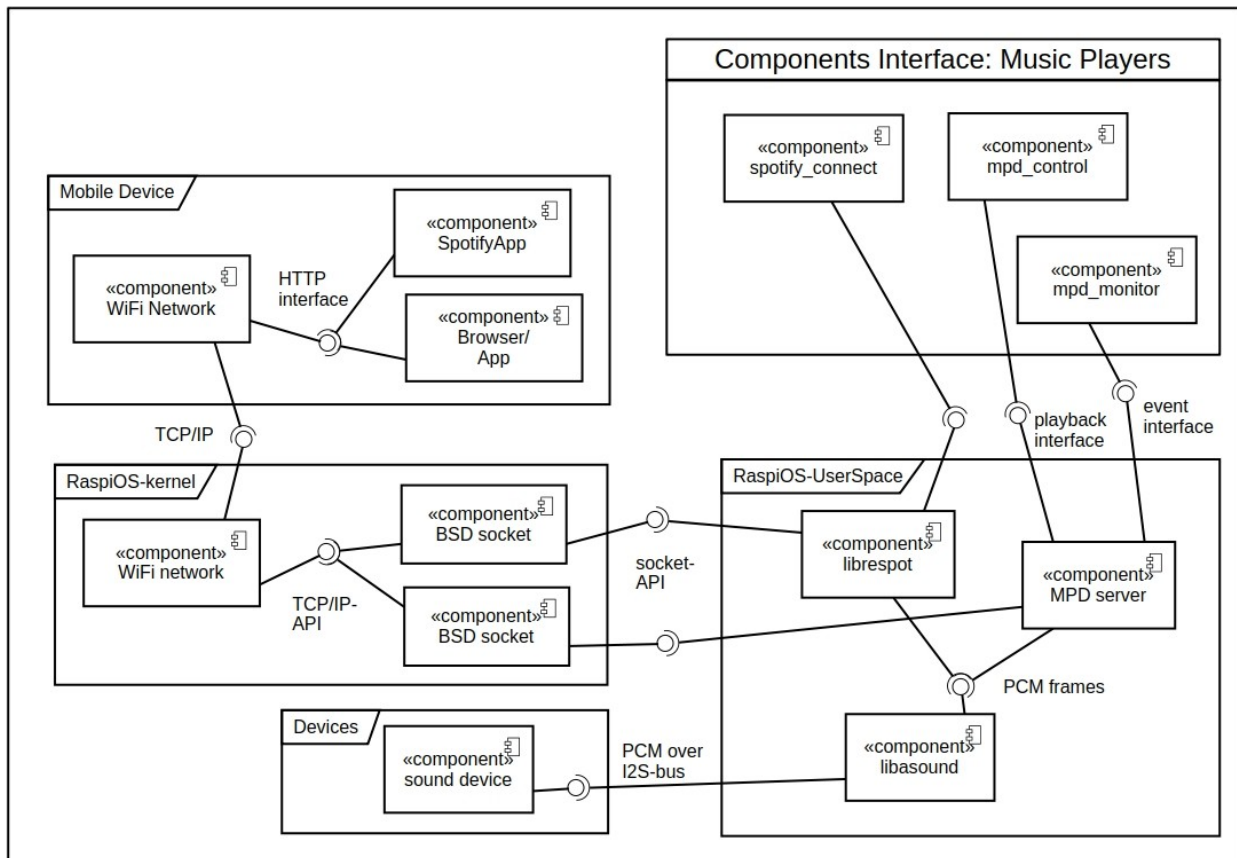
6.2 Messaging (Commands)



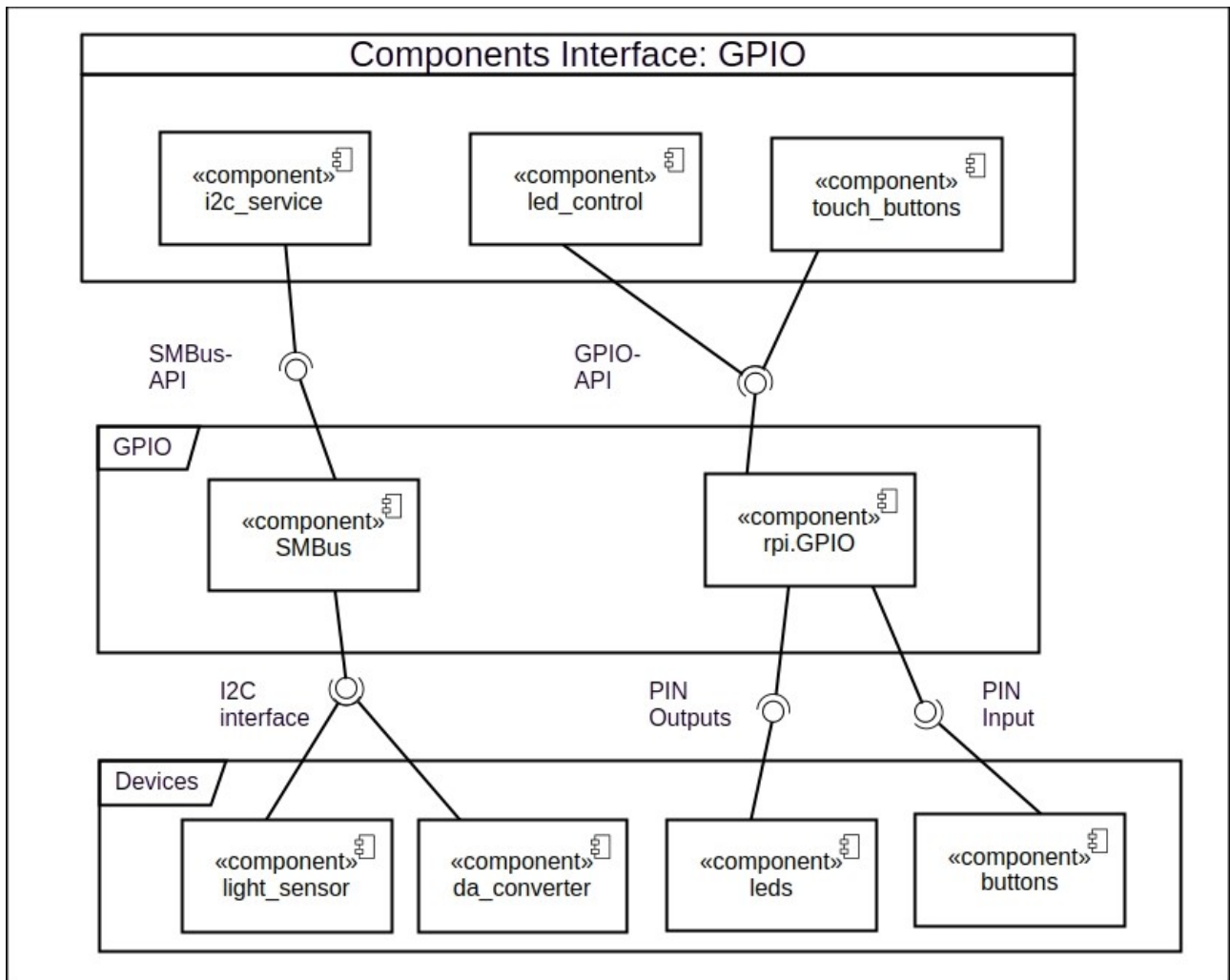
The messaging component interfaces to subscribers are queues. Each subscribers gets an assigned Command queue upon the provided `API:subscribe_commands()`. The Command queue can be monitored by the subscriber to check for new commands.

475 Commands can be published by publishers with the `API:publish_commands()`. The messaging component will distribute the published commands to all subscribers via their assigned queues.

6.3 Music Players

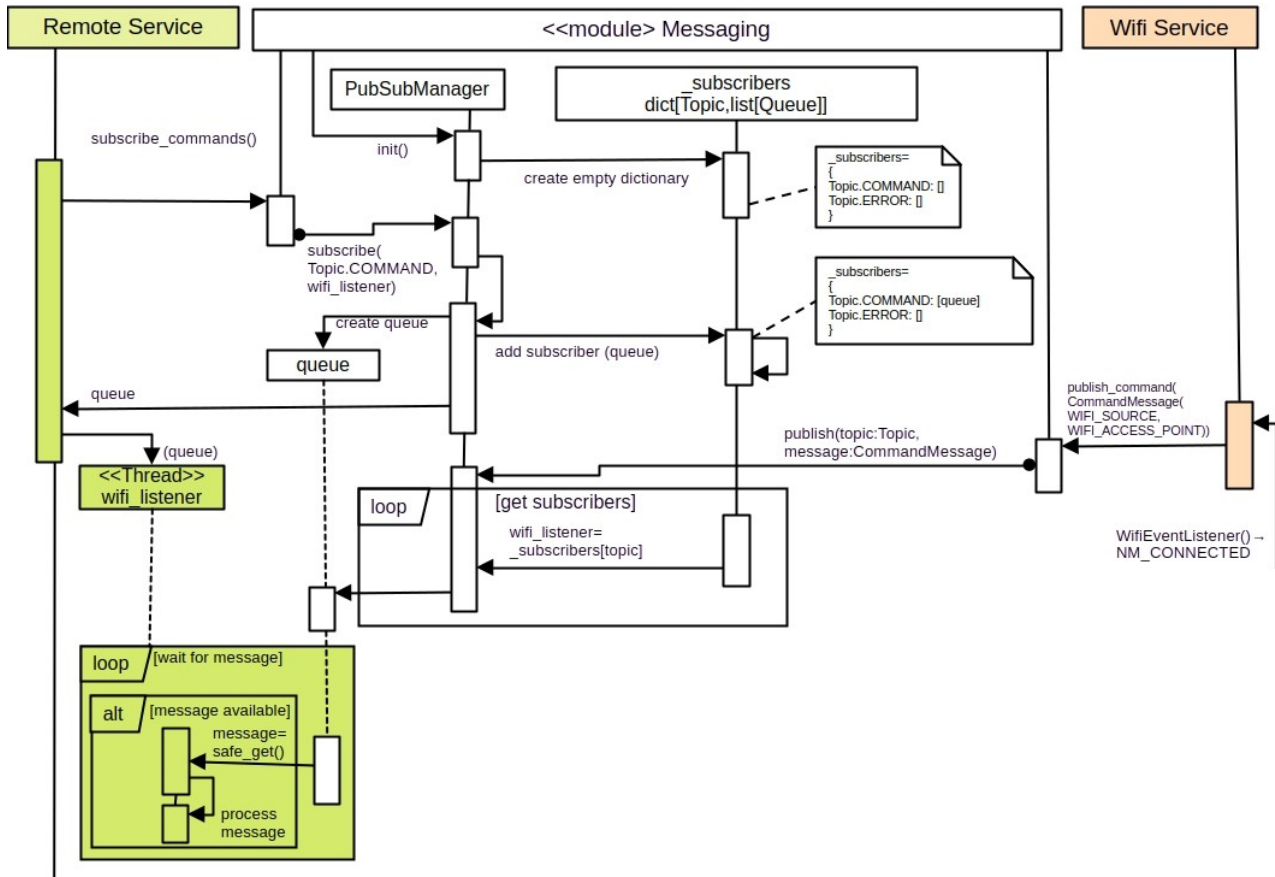


6.4 GPIO components



7.1 Messaging

Messaging: Queue based message subscription



8 The scenario view: use case diagrams

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